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Complete the following definitions.

[2 marks]

The genome is _____

The proteome is _____

Recombinant DNA technology can involve the transfer of fragments of human DNA into bacteria. The bacteria are then used to produce human proteins.

0	4	.	2
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Give **two** reasons why bacteria are able to use human DNA to produce human proteins.

[2 marks]

1 _____

2 _____

0	4	.	3
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Suggest and explain **one** reason why bacteria might **not** be able to produce every human protein.

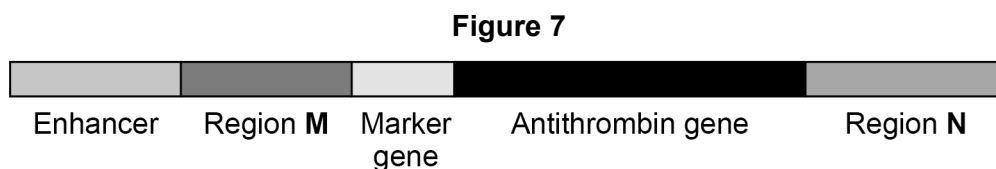
[1 mark]

Antithrombin is a protein. Antithrombin prevents blood from clotting too much.

Some people have a deficiency of antithrombin in their blood, so they need to inject the protein.

Genetically modified goats are used to produce this protein. The human antithrombin gene is transferred into goat embryos. The adult goats then make human antithrombin protein.

Figure 7 shows an example of a DNA fragment that can be transferred into the cells of goats.



0 4 . 4 The enhancer stimulates region **M**.

Name regions **M** and **N** shown in **Figure 7**.

[2 marks]

Region **M** _____

Region **N** _____

0 4 . 5 Explain the purpose of the marker gene.

[1 mark]

0 4 . 6 The enhancer only stimulates region **M** in the milk-producing glands of a goat.

Suggest **two** explanations for the importance of the enhancer being included in the DNA fragment transferred.

[2 marks]

1 _____

2 _____
